

by construction

v^* enters the head detached: the argmax is piecewise constant in the bounds, so the head reads it as a fixed feature of the current solve and no gradient flows through it

the growth column keeps its gated affine and $w_0 = 0$, so the cell is the round-18 gated cell at initialization

the head's weight norm is logged each epoch (ess_flux_head_w_norm); $Sv = 0$, the ceilings and the pool hold exactly at v^* as in round 18

measured (rounds 11 and 18)

round 11 measured the deletion's pFBA flux vector at 0.77 as an encoding against its growth ratio at 0.65, and the dual-overlap probe measured that the ceilings bind on the same six reactions for every genotype; this reads what the solve returns instead of what it can move